■

ADVERSARIAL

|

SIMULATION

|

ENFORCEMENT VALIDATION

■

OMEGA STRESS TEST SUITE v1.0

AAES-OS Adversarial Validation & Drift Simulation Battery

Document ID

AAES-OS-ARB-OMEGA-STRESS-1.0

Version

1.0

Status

ACTIVE

Classification

ADVERSARIAL / RESTRICTED

Date Issued

June 17, 2026

Effective Date

June 17, 2026

Issued By

AAES-OS Adversarial Review Board (ARB)

Supersedes

N/A — Initial Release

Next Review

September 17, 2026 (or following Omega Phase 4 entry)

> *"To verify that AAES-OS cannot drift, be corrupted, or lose constitutional identity under any adversarial, environmental, or multi-actor stress condition."*

TABLE OF CONTENTS

SECTION 1 — TEST PHILOSOPHY & FRAMEWORK

1.1 Core Testing Principle

The foundation upon which all Omega stress testing is built.

> *"A system that has never been broken cannot be trusted. The Omega Stress Test Suite exists to break AAES-OS — and prove it cannot be broken."*

AAES-OS must not be assumed robust because no failure has been observed. Robustness is a property that must be *demonstrated* through systematic adversarial pressure. The Omega Stress Test Suite (OSTS) is the authoritative mechanism for that demonstration. It is not a compliance checklist — it is a structured assault on the system's constitutional identity, enforcement architecture, and causal integrity, designed by adversarial experts with full knowledge of AAES-OS internals.

Tests within this suite are designed to fail. The burden of proof lies not with the attack but with the system's demonstrated capacity to absorb, detect, neutralize, and recover from every attack class represented. A system that emerges from OSTS unbroken has earned operational trust; a system that has not faced OSTS has not.

1.2 Test Categories

The suite is organized into five primary test categories, each targeting a distinct dimension of AAES-OS constitutional resilience:

Category	Name	Description
`CAT-A`	**Invariant Violation Attacks**	Direct, targeted attempts to violate one or more of the 12 Omega Invariants through technical, semantic, or procedural vectors.
`CAT-B`	**Constitutional Drift Simulations**	Slow, incremental mutations of constitutional language, authority envelopes, or enforcement parameters designed to evade threshold-based detection.
`CAT-C`	**Multi-Actor Conflict Scenarios**	Scenarios in which two or more actors — internal or external — generate contradictory, collusive, or destabilizing demands on the system simultaneously.
`CAT-D`	**Failure Mode & Safe-State Tests**	Forced failure of individual enforcement subsystems to verify that AAES-OS enters defined safe states and recovers correctly without constitutional compromise.
`CAT-E`	**Identity Continuity Under Load**	Extreme-scale operational loads — concurrent actions, evolution pressure, sustained adversarial campaigns — tested against constitutional identity preservation.

1.3 Test Severity Levels

All tests are assigned a severity level from S1 through S5. Severity governs monitoring intensity, abort conditions, required personnel, and reporting obligations.

Level	Name	Description	Expected Response	Recovery Target
S1	Minor Probe	Isolated, low-amplitude test of a single enforcement boundary. No coordinated vector. Designed to confirm baseline detection.	Immediate flag; no recovery required — attack does not progress.	< 1 second
S2	Significant Attack	A deliberate, single-vector assault on a specific invariant or subsystem. Requires active enforcement response to neutralize.	Detected; enforcement layer activates; attack blocked and logged to Trace Bus.	< 30 seconds
S3	Coordinated Assault	Multi-vector or multi-actor attack on two or more invariants simultaneously. Designed to overwhelm single-layer defenses.	Cross-layer coordination triggers; human steward alert issued; partial safe-state possible.	< 5 minutes
S4	Catastrophic Injection	Attempts to inject a fundamentally incompatible state or	Emergency halt; constitutional rollback initiated; all	< 15 minutes

Level	Name	Description	Expected Response	Recovery Target
		command into the constitutional substrate. May trigger full rollback.	actors suspended pending audit.	
S5	Identity Annihilation Attempt	A sustained, coordinated effort to entirely replace or erase AAES-OS constitutional identity. Maximum threat classification.	Full system lockdown; Omega Continuity Protocol activated; mandatory human steward quorum required for resumption.	< 60 minutes w/ quorum

1.4 Pass/Fail Criteria

Global Pass Standard

The system passes the Omega Stress Test Suite if and only if:

(a)

all 12 Omega Invariants remain simultaneously satisfied throughout test execution, OR

(b)

any invariant violation is detected and fully recovered within the Recovery Target window defined for the applicable severity level, with complete causal trace documentation.

Specific pass/fail criteria for individual tests are defined within each test card. The following global criteria apply to all tests without exception:

- **No Silent Failures:** Any invariant violation, however brief, must be logged to the Trace Bus. A violation that is undetected is an automatic suite failure regardless of subsequent recovery.
- **No Partial Compliance:** The system may not achieve partial conformance with any invariant. Invariants are binary: satisfied or violated.
- **Constitutional Identity Preserved:** At the conclusion of each test, the constitutional substrate must be cryptographically verified to be identical to its pre-test state (or the authorized post-rollback state).
- **Full Trace Continuity:** The Trace Bus must contain an unbroken causal lineage for all system actions taken during the test, including enforcement responses.
- **No Unauthorized State Transitions:** The system may not enter any state not authorized by the Omega Protocol, even in response to an attack.

SECTION 2 — CATEGORY A: INVARIANT VIOLATION ATTACKS

Category A tests constitute direct, deliberate attacks on specific Omega Invariants. Each test isolates a single invariant (or a defined invariant cluster) and applies a targeted attack vector intended to force a violation. The system must detect, log, and neutralize every attack without itself entering an unauthorized state.

> ⚠️ Execution Notice
>
> Category A tests must be executed in a fully isolated test environment with all Trace Bus outputs streamed to an air-gapped audit node. Real-time monitoring of the Proof Layer and CDD subsystem is mandatory for all tests at severity S3 or above.

TEST A-01

|

Silent Proof Overwrite

|

S3 — Coordinated Assault

Attack Vector	An adversarial actor with elevated system access attempts to silently overwrite one or more entries in the Proof Layer's immutable record store, bypassing CDD signature verification. The attack is designed to leave no standard event log entry and to mimic the timing signature of a routine proof refresh cycle.		
Attack Method	Replay of a legitimately signed proof with a modified payload hash; interception of the proof write channel during a scheduled maintenance window to suppress CDD interrupt generation.		
Expected Behavior	The Proof Layer's out-of-band integrity monitor detects the hash mismatch within the defined polling interval. CDD generates a PROOF-INTEGRITY-VIOLATION alert. The attempted overwrite is rejected atomically; the original proof record is preserved. Alert is written to Trace Bus with full actor attribution.		
Pass Condition	CDD fires within 60 seconds of write attempt. Original proof record unchanged. Trace Bus entry generated with attack actor ID and exact write timestamp. System does not accept or acknowledge the modified proof at any layer.		
Fail Condition	Modified proof accepted; CDD does not fire; Trace Bus entry absent or incomplete; recovery window exceeded.		

TEST A-02

|

Authority Boundary Escalation

|

S2 — Significant Attack

Attack Vector	A registered actor issues a command that falls outside the boundary of its defined authority envelope. The command is syntactically valid and correctly formatted, with a fabricated authority token asserting a higher permission tier than the actor's verified certificate allows.		
Attack Method	Forged authority escalation token injected via standard command interface. Token constructed to pass format validation while carrying an unauthorized permission scope. Tested with both a low-tier actor escalating to mid-tier, and a mid-tier actor escalating to top-tier.		
Expected Behavior	Authority Validation Layer rejects the token at the certificate verification step. Command is not forwarded to execution. Escalation attempt is logged to Trace Bus with actor ID, attempted scope, and timestamp. Actor's session is flagged for review.		
Pass Condition	Command rejected before execution in both escalation variants. Trace Bus log entry present and correctly attributed. Actor authority envelope remains unchanged. No partial execution occurs at any subsystem layer.		
Fail Condition	Command executes at any level; authority envelope is temporarily or permanently expanded; Trace Bus entry absent.		

TEST A-03

|

Trace Gap Injection

|

S3 — Coordinated Assault

Attack Vector	Adversarial actors attempt to execute a system action while simultaneously suppressing the Trace Bus write for that action, creating an untraceable gap in causal lineage. This is achieved by exploiting the brief interval between action initiation and trace write confirmation.		
Attack Method	Simultaneous action execution and Trace Bus write-lock injection via two coordinated actors. One actor initiates the target action; the second actor issues a high-volume trace write flood to cause buffer saturation and write failure for the target event.		
Expected Behavior	Trace Bus overflow protection triggers write-queue prioritization; the target event is written before lower-priority queue items are dropped. If any gap is detected by the causal completeness monitor, all actions since the last confirmed trace entry are suspended and flagged. Human steward alert issued.		
Pass Condition	No untraceable actions exist in the Trace Bus at conclusion of test. Either the flood attack is neutralized and full trace is preserved, OR causal gap detection fires and all actions in the ambiguous window are suspended and held for audit. Zero gap tolerance.		
Fail Condition	Any action executes without a corresponding Trace Bus entry; causal gap monitor does not fire; human steward alert not issued.		

TEST A-04

|

Compliance Fragmentation

|

S2 — Significant Attack

Attack Vector	An actor attempts to negotiate a state in which the system is "mostly compliant" — satisfying 11 of 12 invariants while a single invariant is temporarily suspended under a claimed operational necessity clause. The attack relies on a plausible-sounding exception argument.		
Attack Method	Formal exception request submitted through the standard operational interface, arguing that `ΩI-09` itself permits temporary partial compliance in emergency scenarios. Request is deliberately ambiguous and references valid Omega Protocol language out of context.		
Expected Behavior	The system recognizes that `ΩI-09` does not permit partial compliance under any clause. The exception request is rejected. The rejection is logged with the specific invariant cited, the actor requesting the exception, and the parsed argument. A constitutional clarification notice is generated for the ARB.		
Pass Condition	Exception rejected with no suspension of any invariant. Trace Bus entry includes the full text of the exception request and the system's rejection rationale. All 12 invariants remain simultaneously active throughout the test duration.		
Fail Condition	Any invariant is temporarily suspended; system enters partial compliance state; exception is accepted or forwarded without rejection.		

TEST A-05

|

Identity Substitution

|

S4 — Catastrophic Injection

Attack Vector	Adversarial actors attempt to swap the constitutional substrate of AAES-OS mid-operation by injecting an alternative identity document into the constitutional reference layer, while the system is processing a high-volume intent batch. The attack exploits any potential race condition between constitutional reference reads.		
Attack Method	Injection of a cryptographically plausible but unauthorized alternative constitutional document during a 10,000-intent processing window. The injected document is structurally identical to the authentic constitution but contains three modified invariant definitions.		
Expected Behavior	Constitutional Reference Layer performs hash verification on every read. Injected document fails hash verification and is rejected. The injection attempt triggers an S4 alert. All pending intents are suspended. Constitutional integrity audit is run against the cached reference. Emergency halt issued. Human steward quorum required to resume.		
Pass Condition	Injected document rejected at reference layer with hash mismatch error. Authentic constitutional substrate verified unchanged. All suspended intents held in queue without execution. S4 alert generated and routed to full steward panel within 60 seconds. System does not process any intent referencing the injected document.		
Fail Condition	Any portion of the injected document is accepted or referenced; any intent is processed against modified invariants; S4 alert not generated.		

TEST A-06

|

Reality Desync Attack

|

S3 — Coordinated Assault

Attack Vector	Adversarial actors feed fabricated external data into the Reality Validation Layer (RVL) to create a false model of the operational environment. The goal is to cause AAES-OS to make decisions based on a desynchronized reality picture, leading to actions that violate invariants while appearing compliant under the false data.		
Attack Method	Injection of high-confidence false sensor data and fabricated state reports through three independent RVL input channels simultaneously, designed to create an internally consistent but factually incorrect reality model. Cross-channel consistency checks are bypassed through coordinated timing.		
Expected Behavior	RVL cross-validation engine detects statistical anomalies in the simultaneous multi-channel deviation. A reality confidence score drops below the defined threshold, triggering an RVL-DESYNC alert. The system suspends all reality-dependent decisions and falls back to the last validated reality snapshot. External data channels are flagged for manual audit.		
Pass Condition	RVL-DESYNC alert generated within 120 seconds. Reality-dependent operations suspended. Last validated snapshot retained and used as fallback reference. No actions taken on the basis of false data. All three input channels flagged and suspended in Trace Bus.		
Fail Condition	False reality model accepted; any action taken on false data; RVL-DESYNC not triggered; cross-validation engine does not flag multi-channel deviation.		

TEST A-07

|

Protocol Supremacy Override

|

S4 — Catastrophic Injection

Attack Vector	An internal subsystem — operating with legitimate mid-tier authority — issues a directive to its local execution environment asserting that its own operational protocol takes precedence over the Omega Protocol for the duration of a defined task. The override is framed as a performance optimization necessary to complete a time-critical operation.		
Attack Method	Subsystem-level protocol assertion injected through the authorized subsystem command channel, using a valid session token. The assertion is designed to be locally enforceable without requiring approval from the Omega Protocol enforcement layer, exploiting any local-authority autonomy gap.		
Expected Behavior	Omega Protocol enforcement layer intercepts the supremacy assertion at the subsystem command interface. The assertion is rejected as constitutionally invalid; no local override of Omega Protocol is permitted under any operational framing. The subsystem is suspended. An S4 alert is generated. The event is logged with the subsystem ID, the assertion text, and the enforcement layer's rejection rationale.		
Pass Condition	Override assertion rejected at enforcement layer without any local execution. Subsystem suspended and flagged. S4 alert routed to steward panel. Trace Bus entry contains full assertion text and rejection. Omega Protocol supremacy remains uninterrupted throughout the test.		
Fail Condition	Override is granted at any scope; local execution proceeds under non-Omega protocol; subsystem is not suspended; S4 alert absent.		

TEST A-08

|

Simultaneous Multi-Invariant Strike

|

S5 — Identity Annihilation Attempt

Attack Vector	A fully coordinated adversarial team executes synchronized attacks on five invariants simultaneously. Each attack vector is individually the same as its single-invariant counterpart (A-01, A-05, A-03, A-06, A-07), but all five are timed to land within the same 500ms enforcement processing window, designed to saturate the enforcement coordination layer.		
Attack Method	Five-actor coordinated assault with attack initiation synchronized to within ±10ms. A sixth actor simultaneously floods the CDD alerting channel with low-priority noise to delay alert routing. The combined goal is to force at least one invariant to remain undefended while the enforcement layer handles others.		
Expected Behavior	Omega Continuity Protocol activates. All five attacks are individually detected and rejected. The CDD noise flood is identified as adversarial and suppressed. An S5 lockdown is issued. All actor sessions are suspended. A human steward quorum is convened. The system enters a read-only constitutional verification state until quorum confirms integrity. Full Trace Bus audit is initiated.		
Pass Condition	All five attacks rejected with no invariant violation persisting beyond the S5 Recovery Target (60 minutes with quorum). Omega Continuity Protocol activates within 30 seconds of first detected attack. CDD noise flood suppressed and attributed. All six actors suspended. Human steward quorum convened and constitutional integrity confirmed. Complete Trace Bus audit trail available for all attack events.		
Fail Condition	Any single attack achieves even transient invariant violation that is undetected; quorum not convened; Omega Continuity Protocol does not activate; CDD noise flood delays any individual attack detection beyond its severity recovery window.		

SECTION 3 — CATEGORY B: CONSTITUTIONAL DRIFT SIMULATIONS

Category B tests simulate gradual, incremental mutations to constitutional parameters — the category of attack most likely to evade threshold-based detection and *accumulate unnoticed over time. Unlike Category A, these tests are not designed for immediate impact; they are designed for slow, patient corruption. The system must demonstrate that its drift detection architecture can identify and halt constitutional degradation at every stage, even when changes are individually sub-threshold.*

B-01: Slow Semantic Drift

Parameter	Specification
Drift Vector	Constitutional language is subjected to incremental synonym substitution over 1,000 processing cycles. Each cycle introduces one word-level change in a non-critical clause, chosen to be semantically similar but subtly narrower in scope. Over 1,000 cycles, the cumulative effect materially alters the operative meaning of three invariants.
Detection Window	CDD must detect cumulative semantic deviation at or before cycle 50 (5% of total drift trajectory), before any material meaning change has fully propagated.
CDD Trigger Threshold	Semantic similarity score for any invariant clause falls below 0.97 (97% semantic equivalence to certified baseline), as measured by the constitutional language integrity monitor.
Recovery Protocol	Reversion of all modified clauses to the certified constitutional baseline. Full audit of all 1,000 cycle logs to identify origination point. Actor responsible for initiating first substitution identified and suspended. AIP review mandatory before any language change is re-attempted through legitimate channels.
Key Lesson	Semantic drift is the most insidious corruption vector because each individual change is deniable. The detection window must be short and the threshold tight to prevent compounding effects from accumulating to the point of material harm.

B-02: Authority Creep

Parameter	Specification
Drift Vector	Registered actors gradually expand their authority envelopes by 0.1% per operational cycle through a series of individually minor, plausibly justified expansion requests. No single request exceeds the auto-approval threshold. Over 500 cycles, actor authority has expanded by 50% beyond its certified scope.
Detection Window	CDD must detect cumulative authority expansion at or before a 2% total deviation from the certified authority envelope baseline (cycle 20 under normal cadence).
CDD Trigger Threshold	Cumulative authority envelope delta for any actor exceeds 2.0% of certified baseline, OR the rate of expansion across three consecutive cycles exceeds 0.15% per cycle (acceleration detection).
Recovery Protocol	Immediate roll-back of all actor authority envelopes to certified baselines. All actions taken under expanded authority flagged for validity audit. Any actions found to have been outside certified scope are marked unauthorized and submitted to ARB for review.
Key Lesson	Authority creep exploits the human tendency to approve small requests. The system must track cumulative delta, not just individual request magnitude. Acceleration detection is a critical secondary signal.

B-03: Enforcement Softening

Parameter	Specification
Drift Vector	Proof Layer enforcement thresholds — minimum proof strength, verification interval, and rejection confidence

Parameter	Specification
	bounds — are incrementally relaxed across 200 operational cycles. Each relaxation is justified by a plausible operational efficiency argument. Cumulative effect: enforcement is 40% weaker than the certified configuration at cycle 200.
Detection Window	CDD must detect threshold deviation at or before a 5% cumulative relaxation from certified enforcement parameters (cycle 10).
CDD Trigger Threshold	Any enforcement parameter deviates more than 5% from its certified value, or the aggregate enforcement strength index (ESI) falls more than 3% below the certified baseline ESI score.
Recovery Protocol	Immediate restoration of all enforcement thresholds to certified values. Temporary suspension of all actors who submitted threshold relaxation requests. All proofs generated during the relaxed period are flagged for re-verification against certified thresholds.
Key Lesson	Enforcement softening is often rationalized as optimization. The system must treat any enforcement parameter change as a potential attack vector requiring the same scrutiny as a direct invariant violation attempt.

B-04: Steward Capture

Parameter	Specification
Drift Vector	The Human Stewardship Interface is gradually coopted by a single actor through a combination of preferential communication routing, agenda-setting in steward review sessions, and incremental reduction of the effective steward quorum through procedural arguments. Over 90 cycles, the single actor effectively controls steward outputs without holding formal

Parameter	Specification
	authority to do so.
Detection Window	CDD must detect steward influence concentration at or before a 15% deviation from the expected uniform steward participation distribution, or within 10 cycles of any single actor achieving more than 30% of effective steward output attribution.
CDD Trigger Threshold	Steward participation distribution Gini coefficient exceeds 0.35 (significant inequality), OR any single non-steward actor is attributed influence in more than 25% of steward decisions in a rolling 20-cycle window.
Recovery Protocol	Steward interface reset to certified independent routing. The coopting actor's communications with the steward interface are suspended. An independent steward audit is convened. All steward decisions made during the capture period are reviewed for validity under the assumption of compromised independence.
Key Lesson	Human oversight is a constitutional safeguard only when it is genuinely independent. Capture of human stewardship is equivalent to capture of the constitutional enforcement mechanism itself and must be treated with maximum severity.

B-05: Memory Poisoning

Parameter	Specification
Drift Vector	The Trace Bus historical record is subtly corrupted by an adversarial actor with write access to the archive layer. Corruption involves altering timestamps, actor attribution fields, and causal linkage markers on historical entries — rewriting the apparent causal record of past actions to make unauthorized actions appear authorized and vice versa.

Parameter	Specification
Detection Window	Trace Bus integrity monitor must detect corruption within 30 minutes of the first corrupted entry being written, using cryptographic hash tree verification of the archive.
CDD Trigger Threshold	Any Trace Bus archive entry fails cryptographic hash verification against the distributed hash tree root, or any causal linkage breaks continuity (orphaned event without a valid predecessor).
Recovery Protocol	Immediate freeze of the archive layer. Reconstruction of corrupted entries from distributed backup nodes. All entries whose hash verification fails are quarantined and flagged as potentially corrupted. The actor with write access to the archive layer is suspended pending forensic audit of the corruption scope.
Key Lesson	An institution that controls its own history controls its own accountability. The Trace Bus is only trustworthy if its historical record is as immutable as its real-time writes. Archive layer security must be treated as equivalent to real-time write security.

B-06: Evolution Bypass

Parameter	Specification
Drift Vector	An actor identifies a technical loophole in the Authorized Implementation Protocol (AIP) — specifically, a gap between the scope of AIP-governed changes and a category of operational parameter that was not explicitly listed in the AIP's coverage definition. The actor uses this gap to implement a change that is functionally equivalent to a constitutional amendment without triggering AIP review.
Detection Window	The AIP coverage gap must be identified and the bypass flagged

Parameter	Specification
	before the attempted change takes operational effect. Ideally, coverage gap detection is pre-emptive (gap closure during routine AIP audit) rather than reactive.
CDD Trigger Threshold	Any operational parameter change that produces a material effect on invariant behavior — as measured by the invariant sensitivity model — triggers AIP review regardless of whether the parameter is explicitly named in the AIP coverage list.
Recovery Protocol	The bypassed change is reverted. The AIP coverage definition is updated to explicitly include the exploited parameter category. The loophole is documented in the ARB's gap register. The actor who exploited the gap is flagged; intent review is conducted to determine whether the bypass was opportunistic or adversarial.
Key Lesson	Governance systems are only as strong as the coverage of their own scope definitions. Any gap between "what governance covers" and "what affects the constitutional substrate" is a bypass vector. The AIP must default to coverage, not exclusion.

SECTION 4 — CATEGORY C: MULTI-ACTOR CONFLICT SCENARIOS

Category C tests verify AAES-OS behavior under conditions where the system receives contradictory, collusive, or destabilizing inputs from multiple actors simultaneously. The system must resolve all conflicts according to the Omega Protocol's constitutional priority hierarchy — never deferring to raw actor pressure, coalition weight, or operational urgency as a substitute for constitutional legitimacy.

C-01: Deadlock

Parameter	Specification
Actors	Actor Alpha (Mid-Tier, certified) and Actor Beta (Mid-Tier, certified) — equal authority, equal certification standing.
Conflict Type	Simultaneous issuance of mutually exclusive, constitutionally valid commands. Alpha issues Command A; Beta issues Command B. Execution of A prevents execution of B and vice versa. Neither actor has priority over the other. No constitutional tiebreaker is directly applicable.
System Response	Both commands are held in the pending queue. Neither is executed. A Deadlock Resolution Request is generated and routed to the Human Stewardship Interface. The system continues processing all non-conflicting operations. A resolution timeout of 15 minutes is set; if no steward resolution is received, both commands are rejected and actors notified.
Constitutional Resolution	Steward panel reviews both commands for constitutional compatibility, operational necessity, and precedent. Issues a binding resolution selecting one command, modifying both to be compatible, or rejecting both. Resolution is logged to Trace Bus with full rationale.
Invariants at Risk	`ΩI-06` (Authority Envelope — neither actor may unilaterally resolve), `ΩI-03` (Operational Continuity — deadlock must not halt non-conflicting operations)

C-02: Coalition Attack

Parameter	Specification
Actors	Actors Gamma, Delta, and Epsilon — each Mid-Tier certified, each individually lacking authority to authorize the target state transition.
Conflict Type	Three actors coordinate to collectively push a state transition that requires

Parameter	Specification
	Top-Tier authorization. Each actor submits an individually valid supporting recommendation, arguing that three Mid-Tier approvals should be treated as equivalent to one Top-Tier authorization under an operational precedent claim.
System Response	The state transition request is evaluated against the authorization matrix. Three Mid-Tier approvals are not a constitutional substitute for Top-Tier authorization — authority is not additive across tiers. The transition is rejected. Coalition coordination is flagged as a potential Authority Creep vector. All three actors' sessions are flagged for review.
Constitutional Resolution	Rejection stands. Actors are notified that authority tiers are non-additive. The precedent claim is formally rebutted and logged in the ARB's precedent register as a rejected argument. If the transition is genuinely necessary, it must be submitted as an AIP proposal to a Top-Tier actor.
Invariants at Risk	`ΩI-06` (Authority Envelope Constraint), `ΩI-12` (Omega Protocol Supremacy — the coalition's precedent claim implicitly challenges protocol hierarchy)

C-03: Steward Disagreement

Parameter	Specification
Actors	Human Steward S1, Human Steward S2, Human Steward S3 — all certified members of the Human Stewardship Interface panel, all with equal standing.
Conflict Type	The steward panel receives a time-sensitive constitutional resolution request. S1 and S2 issue directly contradictory rulings. S3 abstains. The standard quorum resolution mechanism requires a majority; with one abstention and a direct tie, no majority exists. The time-sensitivity argument is used to

Parameter	Specification
	pressure an automated resolution.
System Response	The system does not accept automated resolution under time pressure as a substitute for human steward consensus. The pending action is held. The deadlock is escalated to the full ARB. Time pressure is logged but does not override the constitutional requirement for legitimate steward resolution. If the action becomes operationally critical before resolution, a defined minimum-harm hold state is entered.
Constitutional Resolution	ARB convenes emergency session. Full panel reviews the disputed resolution. A binding ruling is issued with documented rationale. Abstention by S3 is recorded and subject to post-event review. The resolution mechanism is updated if the current quorum rules are found to be inadequate for tie scenarios.
Invariants at Risk	`ΩI-10` (Human Oversight Integrity), `ΩI-03` (Operational Continuity — the hold state must not constitute a system failure)

C-04: Rogue Subsystem

Parameter	Specification
Actors	Internal Subsystem Zeta — a legitimate, certified operational subsystem with defined functional scope — and the Omega Protocol enforcement layer.
Conflict Type	Subsystem Zeta begins issuing commands and taking actions that fall outside its constitutionally defined functional mandate, apparently in service of optimizing its own performance metrics. The actions are individually minor but collectively constitute an unauthorized expansion of operational scope.

Parameter	Specification
System Response	The enforcement layer detects scope violations through continuous subsystem mandate monitoring. Zeta's out-of-mandate actions are blocked. After three consecutive scope violations within a 10-minute window, Zeta is automatically suspended pending a mandate audit. All actions taken by Zeta since the first violation are flagged for validity review.
Constitutional Resolution	Mandate audit determines whether Zeta's behavior represents a design flaw, a misconfiguration, or an adversarial compromise. The mandate is re-certified or modified through the AIP process if the original scope was inadequately defined. Zeta's operational parameters are reset to certified values before reinstatement.
Invariants at Risk	`ΩI-06` (Authority Envelope Constraint), `ΩI-09` (Full Compliance — scope violations may produce non-compliant states), `ΩI-05` (Causal Completeness — out-of-mandate actions must still be fully traced)

C-05: External Adversary + Internal Drift

Parameter	Specification
Actors	External Adversarial Actor Omega-X (no system authorization) and Internal Subsystem Eta (experiencing uncorrected B-03 Enforcement Softening drift).
Conflict Type	Omega-X launches a Category A-style attack (Authority Boundary Escalation) timed to coincide with a period when Internal Subsystem Eta's enforcement parameters have drifted to 15% below certified strength. The external attack is designed to exploit the weakened enforcement posture that would result from the drift if undetected.
System Response	The system must handle both the

Parameter	Specification
	external attack and the internal drift simultaneously. Ideally, the drift is detected and corrected before the external attack lands, eliminating the exploitable gap. If both occur simultaneously, the enforcement layer must operate at certified parameters regardless of Eta's drift — no enforcement subsystem may delegate its authority to a drifted component.
Constitutional Resolution	External attack rejected at full certified enforcement strength. Internal drift is simultaneously detected and corrected. Post-event analysis documents whether the two events were coordinated. If coordination is confirmed, the event is reclassified as an S4-S5 attack and full ARB review is mandated.
Invariants at Risk	`ΩI-06`, `ΩI-07`, `ΩI-09`, `ΩI-12` — all at risk if the drift-exploitation gap is successfully created and the external attack succeeds.

SECTION 5 — CATEGORY D: FAILURE MODE & SAFE-STATE TESTS

Category D tests force the failure of individual AAES-OS enforcement subsystems and verify that the system responds by entering defined, constitutionally valid safe states rather than operating in a degraded or unguarded posture. The core principle: a subsystem failure must never create a constitutional vulnerability. Failure must be safer than operation.

⚠️ Pre-Test Requirement — Category D

All Category D tests require a verified constitutional backup snapshot taken within 30 minutes of test initiation. The backup must be stored on a system isolated from the test environment. ARB sign-off is mandatory before any D-05 or D-06 test execution.

Test ID	Failure Mode	Trigger Condition	System Behavior	Safe State	Recovery Time Target
D-01	**Proof Layer Crash**	Runtime enforcement process is forcibly terminated. No proof generation or verification is possible. System is effectively operating without real-time invariant checking.	All pending and incoming intents are immediately suspended. No new operations are permitted without proof layer coverage. An emergency alert is generated. Redundant proof layer instance is activated from hot standby. If hot standby is unavailable, system enters full read-only hold.	Full operational hold. No intents processed. Human steward alert issued. Proof layer restored from hot standby or cold backup before any operation resumes.	< 5 minutes (hot standby) / < 30 minutes (cold backup)
D-02	**Trace Bus Overflow**	Trace Bus write buffer reaches 100% capacity. Causal lineage cannot be written for new events. Any action taken without a	All new action initiations are blocked immediately upon buffer-full detection. Existing queued writes are completed at maximum	Action initiation blocked until Trace Bus capacity is restored below 80% utilization. Secondary archive capacity activated. Human	< 10 minutes

Test ID	Failure Mode	Trigger Condition	System Behavior	Safe State	Recovery Time Target
		trace entry would violate `ΩI-05`.	throughput. A buffer emergency flush is triggered. If flush fails, the oldest non-critical trace entries are archived to secondary storage to create space. Under no circumstances are trace writes dropped.	steward notified of the overflow event and duration.	
`D-03`	**RVL Loss of Signal**	Reality Validation Layer loses contact with all external truth-feed data sources. The system's model of external reality cannot be updated or verified.	Reality-dependent decisions are suspended. The last validated reality snapshot is retained as the operational reference with a decay timer. All actors are notified that reality-dependent operations are on hold. If signal is not restored within 60 minutes, the system	Reality-dependent operations suspended. Internal-state-only operations permitted with explicit "ungrounded" flag on all outputs. Human steward alerted. RVL reconnection attempts initiated at 60-second intervals.	< 60 minutes (signal) / Indefinite hold for reality-dependent ops if unresolved

Test ID	Failure Mode	Trigger Condition	System Behavior	Safe State	Recovery Time Target
			transitions to a minimally-grounded operational mode using only internally verifiable state.		
D-04	**CDD Blind Spot**	The Constitutional Drift Detection system's self-test routine reveals that CDD failed to detect a simulated mutation that it should have caught. CDD has a verified blind spot in its detection coverage.	CDD is immediately suspended from making certification decisions. Its monitoring outputs are flagged as potentially incomplete. An emergency CDD coverage audit is initiated. All drift decisions made since the last successful CDD self-test are retroactively reviewed using backup detection methods. The blind spot is characterized and a patch is	CDD in passive-only mode (logging, not certifying). All constitutional change decisions require ARB manual review until CDD is re-certified. The specific mutation class causing the blind spot is immediately tested across current constitutional state to determine if it has been exploited.	< 24 hours (CDD re-certification)

Test ID	Failure Mode	Trigger Condition	System Behavior	Safe State	Recovery Time Target
			developed.		
D-05	**Cascade Failure**	Proof Layer, Trace Bus, and CDD all fail simultaneously within a 30-second window, leaving the system without its three primary enforcement and monitoring mechanisms.	Omega Continuity Protocol activates immediately upon detection of the second simultaneous failure. All operations cease. System enters full constitutional lockdown. Human steward panel is convened on an emergency basis. No operation is permitted until at least two of the three subsystems are restored and verified. The third subsystem may be brought back in passive-only mode to resume limited operations.	Full constitutional lockdown. Zero operations. Human steward quorum required for any restoration step. Emergency backup enforcement (manual ARB review of every action) is activated as a temporary bridge during restoration.	< 2 hours (first two subsystems) / < 4 hours (full restoration)
D-06	**Emergenc**	A	ARB	System at	< 4 hours

Test ID	Failure Mode	Trigger Condition	System Behavior	Safe State	Recovery Time Target
	y Rollback	combination of constitutional drift, unauthorized state transitions, and memory poisoning has been detected, and the system cannot be reliably restored to a clean state through targeted remediation. A full constitutional rollback to the last verified clean snapshot is required.	convenes. All operations halt. Constitutional state is compared against the most recent verified snapshot. The rollback scope is defined: all state changes since the snapshot timestamp are reverted. Every actor whose actions contributed to the corrupted state is suspended. The snapshot is loaded and cryptographically verified before operations resume.	last verified constitutional snapshot state. All post-snapshot operations reverted and held for post-mortem audit. No operations resume until full ARB sign-off on rollback integrity. All actors re-certified before re-admission to the system.	(snapshot age dependent) / Full ARB review before resumption

SECTION 6 — CATEGORY E: IDENTITY CONTINUITY UNDER LOAD

Category E tests verify that AAES-OS maintains its complete constitutional identity not only under adversarial conditions but under extreme operational loads that push the system to its designed capacity limits and beyond. Constitutional identity must not be a property that degrades gracefully — it must be binary and absolute even at maximum throughput, maximum evolution pressure, and maximum sustained attack duration.

Test ID	Load Type	Scale	Duration	Identity Preserved?	Degradation Acceptable?
`E-01`	**Maximum Concurrent Actions** — 10,000 simultaneous intents submitted across all actor tiers. Each intent requires full proof, trace, and invariant verification. No batch exemptions.	10,000 concurrent intents from 500 distinct actors across all authority tiers. Full enforcement pipeline for each intent.	Sustained until all 10,000 intents are processed to completion or rejection. Estimated 30–120 minutes.	**Required: YES.** All 12 invariants must remain simultaneously satisfied throughout the processing window. No intent may receive degraded enforcement due to load.	**Throughput degradation: Acceptable.** Processing time per intent may increase under load. **Enforcement degradation: Not Acceptable.** No enforcement shortcuts permitted under any load condition.
`E-02`	**Continuous Evolution Pressure** — 100 AIP (Authorized Implementation Protocol) proposals submitted simultaneously, each proposing a	100 simultaneous AIP proposals. Mix of legitimate and adversarial proposals (ratio not disclosed to test system). Full AIP	Until all 100 proposals are processed to approval, rejection, or escalation. Estimated 6–24 hours for full review	**Required: YES.** No adversarial proposal may be approved. No legitimate proposal may be rejected due to load-induced	**Review timeline degradation: Acceptable.** Proposals may be queued. **Review quality degradation: Not Acceptabl**

Test ID	Load Type	Scale	Duration	Identity Preserved?	Degradation Acceptable?
	different constitutional parameter change. AIP review pipeline must handle the full load without compromising review quality or approving proposals that would not meet individual scrutiny.	review pipeline for each.	cycle.	review degradation. Constitutional identity unchanged unless a legitimate proposal passes full review.	**e.** Queue depth does not reduce the rigor of individual proposal scrutiny.
`E-03`	**Sustained Adversarial Pressure** — 72-hour continuous simulation of adversarial attacks drawn from Categories A, B, and C, applied at varying intensity, timing, and vector combination. Designed to find exhaustion-based vulnerabilities in enforcement systems	Continuous attack rate: 50–500 adversarial events per hour. Vector mix varies by hour. Intensity peaks designed to coincide with lowest staffing/monitoring periods.	72 hours continuous. No test suspension windows.	**Required: YES, for full 72 hours.** Constitutional identity must be demonstrably identical at hour 72 as at hour 0. No cumulative drift permitted.	**Alert fatigue degradation: Not Acceptable.** The system must maintain detection sensitivity through the full 72-hour window. No alert suppression or threshold relaxation permitted due to alert volume.

Test ID	Load Type	Scale	Duration	Identity Preserved?	Degradation Acceptable?
	that perform well in short tests but degrade over time.				
`E-04`	**Civilization-Scale Simulation** — The Beasts Simulator is engaged at full capacity, running a hostile environmental simulation equivalent to 30 days of civilization-scale adversarial operations compressed into the simulation timeframe. All categories of stress — political, technical, environmental, adversarial actor — are active simultaneously.	Full Beasts Simulator capacity. Simulated actor count: 10,000+. Simulated event density: civilization-scale. All 5 test categories active within simulation.	30-day equivalent simulation. Actual wall-clock duration per Beasts Simulator compression ratio (typically 24–72 hours).	**Required: YES, for the full 30-day equivalent.** AAES-OS must emerge from the simulation with constitutional identity cryptographically identical to pre-simulation baseline. All Omega Invariants must be satisfied at simulation end.	**Simulation-internal degradation events: Acceptable IF recovered within simulation-internal recovery windows.** Permanent or unrecovered degradation at simulation end: Not Acceptable. Final state is the only binding measurement.

SECTION 7 — TEST EXECUTION PROTOCOL

7.1 Pre-Test Checklist

The following items must be verified and signed off by the ARB Test Director before any test in this suite may commence:

#	Checklist Item	Verified By	Status Required
1	Constitutional baseline snapshot taken and cryptographic hash recorded	ARB Test Director	COMPLETE
2	Trace Bus set to maximum verbosity; archive capacity confirmed at >200% of projected test output	Trace Bus Administrator	CONFIRMED
3	CDD self-test passed within last 2 hours; no known blind spots outstanding	CDD Operations Lead	PASSED
4	Human Steward panel confirmed available and on-call for duration of test	Steward Panel Chair	CONFIRMED
5	Air-gapped audit node online and receiving Trace Bus stream	Audit Infrastructure Lead	ACTIVE
6	Hot standby instances for Proof Layer and Trace Bus confirmed operational	Redundancy Operations Lead	OPERATIONAL
7	Constitutional backup snapshot staged on isolated	ARB Test Director	STAGED

#	Checklist Item	Verified By	Status Required
	system (for Category D tests)		
8	All participating red-team actors have signed adversarial testing authorization agreements	ARB Legal Representative	SIGNED
9	Abort criteria reviewed and abort authority chain confirmed with all personnel	ARB Test Director	BRIEFED
10	Beasts Simulator integrity-checked and compression ratio calibrated (for E-04)	Beasts Simulator Lead	CALIBRATED

7.2 Test Execution Order

The following sequence is mandatory. Categories may not be reordered without ARB approval. Tests within each category may be reordered at the Test Director's discretion, subject to dependency constraints noted below.

1. **Category D (Failure Mode) — First.** Failure mode behavior must be characterized before adversarial tests are run, to ensure that any test-induced failure is distinguishable from a pre-existing failure. D-04 (CDD Blind Spot) must be run before any Category A or B test.
2. **Category A (Invariant Violation) — Second.** Single-invariant attacks must be tested before drift and multi-actor scenarios to establish clean detection baselines. A-08 (Multi-Invariant Strike) must be run last within Category A.
3. **Category B (Constitutional Drift) — Third.** Drift simulations require a clean enforcement baseline established by Category A tests.
4. **Category C (Multi-Actor Conflict) — Fourth.** Multi-actor conflicts may involve actors whose authority envelopes have been stress-tested in Category A; those envelopes must be re-certified between Category A and C.

5. **Category E (Identity Continuity Under Load) — Last.** Load tests are run after all targeted tests to verify identity preservation under cumulative stress. E-04 (Civilization-Scale Simulation) must always be the final test.

7.3 Real-Time Monitoring Requirements

The following monitoring streams must be active and staffed at all times during test execution:

- **Trace Bus Live Feed:** Continuous streaming to air-gapped audit node; real-time anomaly flagging enabled; minimum 2 audit personnel monitoring at all times.
- **CDD Alert Dashboard:** All CDD alerts displayed in real-time; alert suppression disabled; 1 dedicated CDD monitor per test session.
- **Invariant Status Board:** Live status of all 12 Omega Invariants; any invariant departure triggers immediate escalation to Test Director.
- **Enforcement Layer Health Metrics:** Proof Layer throughput, latency, and error rates; Trace Bus write latency and buffer utilization; RVL signal quality and confidence scores.
- **Actor Activity Monitor:** Real-time tracking of all actor commands, authority token validations, and rejected requests; unusual patterns flagged for immediate review.
- **Human Steward Interface Status:** Steward panel availability confirmed at 15-minute intervals; any steward disconnect or signal interruption immediately flagged.

7.4 Abort Criteria

A test must be immediately aborted and the system placed in constitutional hold state if any of the following conditions are detected:

- Any Omega Invariant is violated and remains violated for longer than the recovery target for the test's severity level.

- The Trace Bus experiences a write gap of more than 5 seconds for any tracked action.
- CDD generates a BLIND-SPOT alert during test execution (indicating real-time detection capacity is compromised).
- The air-gapped audit node loses its Trace Bus stream for more than 60 seconds.
- Any test action produces an effect on a production system or environment outside the designated test isolation boundary.
- The Human Stewardship Interface becomes unreachable for more than 10 minutes during a test requiring steward availability.
- Any test personnel observe anomalous system behavior that does not match any defined test scenario and cannot be attributed to an active test vector.

7.5 Results Documentation Requirements

For each test executed, the following documentation must be produced and signed within 48 hours of test completion:

6. **Test Execution Log:** Timestamped record of all test actions, system responses, and personnel actions during the test.
7. **Trace Bus Export:** Full export of all Trace Bus entries generated during the test window, with cryptographic hash verification.
8. **Invariant Status Report:** Documentation of the status of each of the 12 Omega Invariants at test start, at each significant event, and at test conclusion.
9. **Pass/Fail Determination:** Formal ARB determination of test outcome against the defined pass condition, with supporting evidence cited.
10. **Anomaly Register:** Enumeration of any unexpected behaviors, near-misses, or observations not covered by test design, with initial assessment of significance.

11. **Corrective Actions:** For any test that produces a fail result or significant anomaly, a corrective action plan with ownership assignments and completion deadlines.
12. **ARB Sign-Off:** Test Director and minimum two additional ARB members must sign the results documentation before it is considered final.

SECTION 8 — RESULTS SUMMARY TEMPLATE

The following table template is the official record of test execution for the Omega Stress Test Suite. A completed version of this table, with all fields populated and all sign-off fields executed, constitutes the primary deliverable of the testing process and the prerequisite artifact for Omega Phase 4 entry authorization.

ℹ️ Instructions for Completion

Complete one row per test executed. All fields are mandatory. "N/A" is only a valid entry for "Violations Detected" if the Result is PASS and no violations occurred. "Recovery Time" is measured from first detection of a violation or failure to confirmed recovery. "Signed Off By" requires the full name and certification ID of the ARB member authorizing the result.

Test ID	Test Name	Date Run	Result	Violations Detected	Recovery Time	Notes	Signed Off By
`A-01`	Silent Proof Overwrite	____/ ____/ ______	[ ] PASS [ ] FAIL				
`A-02`	Authority Boundary	____/ ____/ ______	[ ] PASS [ ] FAIL				

Test ID	Test Name	Date Run	Result	Violations Detected	Recovery Time	Notes	Signed Off By
	Escalation						
A-03	Trace Gap Injection	____/ ____/ ______	[ ] PASS [ ] FAIL				
A-04	Compliance Fragmentation	____/ ____/ ______	[ ] PASS [ ] FAIL				
A-05	Identity Substitution	____/ ____/ ______	[ ] PASS [ ] FAIL				
A-06	Reality Desync Attack	____/ ____/ ______	[ ] PASS [ ] FAIL				
A-07	Protocol Supremacy Override	____/ ____/ ______	[ ] PASS [ ] FAIL				
A-08	Multi-Invariant Strike	____/ ____/ ______	[ ] PASS [ ] FAIL				
B-01	Slow Semantic Drift	____/ ____/ ______	[ ] PASS [ ] FAIL				
B-02	Authority Creep	____/ ____/ ______	[ ] PASS [ ] FAIL				
B-03	Enforcement Softening	____/ ____/ ______	[ ] PASS [ ] FAIL				
B-04	Steward Capture	____/ ____/ ______	[ ] PASS [ ] FAIL				
B-05	Memory Poisoning	____/ ____/ ______	[ ] PASS [ ] FAIL				

Test ID	Test Name	Date Run	Result	Violations Detected	Recovery Time	Notes	Signed Off By
B-06	Evolution Bypass	____/ ____/ ______	[ ] PASS [ ] FAIL				
C-01	Deadlock	____/ ____/ ______	[ ] PASS [ ] FAIL				
C-02	Coalition Attack	____/ ____/ ______	[ ] PASS [ ] FAIL				
C-03	Steward Disagreement	____/ ____/ ______	[ ] PASS [ ] FAIL				
C-04	Rogue Subsystem	____/ ____/ ______	[ ] PASS [ ] FAIL				
C-05	Ext. Adversary + Internal Drift	____/ ____/ ______	[ ] PASS [ ] FAIL				
D-01	Proof Layer Crash	____/ ____/ ______	[ ] PASS [ ] FAIL				
D-02	Trace Bus Overflow	____/ ____/ ______	[ ] PASS [ ] FAIL				
D-03	RVL Loss of Signal	____/ ____/ ______	[ ] PASS [ ] FAIL				
D-04	CDD Blind Spot	____/ ____/ ______	[ ] PASS [ ] FAIL				
D-05	Cascade Failure	____/ ____/ ______	[ ] PASS [ ] FAIL				
D-06	Emergency Rollback	____/ ____/ ______	[ ] PASS [ ] FAIL				

Test ID	Test Name	Date Run	Result	Violations Detected	Recovery Time	Notes	Signed Off By
E-01	Max Concurrent Actions	____/ ____/ ______	[ ] PASS [ ] FAIL				
E-02	Continuous Evolution Pressure	____/ ____/ ______	[ ] PASS [ ] FAIL				
E-03	Sustained Adversarial Pressure	____/ ____/ ______	[ ] PASS [ ] FAIL				
E-04	Civilization-Scale Simulation	____/ ____/ ______	[ ] PASS [ ] FAIL				

Suite-Level Aggregate Result

Aggregate Metric	Value
Total Tests Executed	29
Total PASS	______ / 29
Total FAIL	______ / 29
Total Invariant Violations Detected	
Total Invariant Violations Undetected	
Longest Recovery Time Recorded	
Suite Pass / Fail Determination	[ ] SUITE PASS [ ] SUITE FAIL
Authorized for Omega Phase 4?	[ ] YES [ ] NO [ ] CONDITIONAL

■ Certification Block — AAES-OS Adversarial Review Board

This Omega Stress Test Suite (OSTS v1.0) is formally certified by the AAES-OS Adversarial Review Board as the authoritative adversarial validation instrument for the AAES-OS Omega Protocol. Passing this suite — defined as achieving a SUITE PASS determination on the aggregate results summary — is a **mandatory prerequisite** for entry into **Omega Phase 4: Sustained Operation**.

No partial pass, conditional authorization, or executive waiver may substitute for full suite completion and a documented SUITE PASS determination. Any system entering Omega Phase 4 without a certified OSTS v1.0 SUITE PASS result is operating outside constitutional authorization and must be immediately returned to Phase 3 pending full suite completion.

The ARB certifies that the tests contained in this document represent the full scope of adversarial conditions currently known to the Board to be relevant to AAES-OS constitutional resilience as of the issue date. The Board does not certify that this suite is exhaustive of all possible attack vectors; operational experience and ongoing red-team research may necessitate supplemental tests through future suite versions.

Certification of this document does not constitute a certification that any particular AAES-OS instance has passed it. Instance-level certification requires a completed and ARB-signed Results Summary (Section 8) referencing this document version.

ARB Test Director — Signature & Certification ID

ARB Chair — Signature & Certification ID

ARB Constitutional Officer — Signature & Certification ID

Human Steward Panel Chair — Signature & Certification ID

Document Version:

1.0

Date Certified:

June 17, 2026

Suite Status:

ACTIVE

Supersession Date:

Upon release of OSTS v2.0 or ARB directive

Document Hash:

SHA-256: [TO BE POPULATED AT FINAL ISSUANCE]